

Brain Tumor MRI Classification with Grad-CAM-Guided Training: Explainability, Tumor-Region Ensembles and Leakage-free Evaluation

Technical Report

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Abstract

We classify brain tumors (glioma, meningioma, pituitary tumor) on the Figshare contrast-enhanced T1 MRI dataset under the official patient-wise 5-fold protocol. Grad-CAM, scored against the ground-truth tumor masks, shows that fine-tuned CNNs are mostly right for the wrong reasons: only 3–9 % of their Grad-CAM mass lies on the tumor. We turn the masks into a training signal that penalises Grad-CAM mass outside the (dilated) tumor. This raises the pointing-game score of EfficientNet-B3 from 0.26 to 0.91 and its accuracy by 1.0–1.3 points, without needing masks at test time. We further add a tumor-region classifier fed with out-of-fold Mask R-CNN crops. We select an ensemble on validation folds only and refit it on 80 % of the data. The final ensemble reaches 97.64 ± 1.12 % test accuracy, above the closest published result on the same protocol (97.3 %) and on par with the best (98 %). Bigger backbones, EMA weights and higher input resolution give no gain. Mask R-CNN reaches mean Dice 0.76 under the same protocol.

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1 Introduction

Public brain-tumor MRI datasets make it easy to report accuracies above 95 %. Two questions are often left open: is the evaluation free of patient leakage, and does the network base its decision on the tumor at all? This report answers both for the Figshare dataset [1, 2] and uses the answer to the second question to improve the classifiers.

Motivation. An earlier version of this project split all slices at random (70/15/15). With about 13 near-identical slices per patient this puts the same patient in training and test. The same InceptionV3 drops from 97.7 % to 92.8 % when patients are separated. All results in this report therefore use the official patient-wise folds, the protocol of the published works we compare against [2, 4, 5].

Contributions.

1. A quantitative Grad-CAM analysis against ground-truth tumor masks that shows weak localisation and a dataset shortcut (Section 4).
2. Grad-CAM-guided training, which makes the networks look at the tumor and improves their accuracy and stability (Section 5).
3. A tumor-region classifier on out-of-fold Mask R-CNN crops and a validation-selected, refit ensemble that matches the best published accuracy on this protocol (Section 7).
4. Ablations of recipe, backbone, EMA, resolution, crop context, ensemble composition and training-set size (Section 6).

2 Data and protocol

The Figshare brain tumor dataset [1, 2] contains 3064 contrast-enhanced T1 slices (512×512) from 233 patients: 1426 glioma, 708 meningioma and 930 pituitary tumor slices. Each `.mat` file stores the image, the patient ID, the label and a manually delineated binary tumor mask. The dataset ships with `cvind.mat`, an official patient-wise assignment of every slice to one of five folds. Slices are min-max scaled to 8 bit (the formula of the Figshare README) and replicated to three channels. For the secondary 4-class experiments we add the 395 *no tumor* images of the Kaggle *Brain Tumor Classification (MRI)* dataset [3], each assigned to a fold with a fixed seed.

Development runs. For test fold k , fold $k+1$ is the validation fold and the remaining three folds are used for training (60 % of the data). The best epoch, the configurations and the ensemble are chosen on *pooled validation accuracy*: each image is a validation image in exactly one fold.

Refit. The selected configurations are retrained on all four non-test folds (80 %, as in the published works) for the fixed 30-epoch schedule, and the final weights are evaluated once on the test fold. We report mean $\pm$ standard deviation of test accuracy over the five folds and the pooled accuracy over all 3064 slices.

3 Methods

3.1 Classifiers and training recipe

We fine-tune five ImageNet-pretrained CNNs: InceptionV3 [7], EfficientNet-B3 [8], EfficientNetV2-S [9], ConvNeXt-T [10] and, for the Grad-CAM analysis, AlexNet [6]. Two training recipes are compared:

Table 1: Training recipes.

Recipe	Settings
baseline	resize (224; InceptionV3 299) + ImageNet normalisation; AdamW, lr 10^{-4} , weight decay 10^{-4} , batch 128, 15 epochs, cross-entropy; best epoch by validation accuracy
improved	MRI-plausible augmentation (horizontal flip; affine: scale $\pm 10\%$, shift $\pm 5\%$, rotation $\pm 15^\circ$; brightness/contrast $\pm 15\%$; gamma); AdamW, lr $2 \cdot 10^{-4}$, weight decay 0.05, batch 32, 30 epochs, 1-epoch warm-up + cosine; label smoothing 0.1; InceptionV3 auxiliary loss (0.4); EfficientNet-B3 at its native 300×300 ; bf16; horizontal-flip test-time augmentation; best epoch by validation macro-F1

3.2 Grad-CAM and localisation metrics

For an input x and class c , let $A^k \in \mathbb{R}^{h \times w}$ be the k -th feature map of the last convolutional block. Grad-CAM [12] is

$$\text{CAM}_c = \text{ReLU}\left(\sum_k \alpha_k^c A^k\right), \quad \alpha_k^c = \frac{1}{hw} \sum_{i,j} \frac{\partial y_c}{\partial A_{ij}^k}, \quad (1)$$

bilinearly upsampled to the input size. For every test image we compute the Grad-CAM of the predicted class and, given the tumor mask M :

- **Pointing game** [13]: 1 if $\arg \max \text{CAM}$ lies in M dilated by 7% of the image side.
- **Energy in tumor**: $\sum \text{CAM} \odot M / \sum \text{CAM}$. A uniform map scores the tumor area, 1.7% on average (chance level).
- **CAM-IoU**: IoU between $\{\text{CAM} > 0.5 \max \text{CAM}\}$ and M .
- **Energy in brain**: share of CAM mass inside the head (largest Otsu foreground component), a shortcut indicator.

3.3 Grad-CAM-guided training

Following the “right for the right reasons” idea [14], we ask the Grad-CAM of the true class y to fall on the tumor:

$$\mathcal{L} = \mathcal{L}_{\text{CE}} + \lambda \frac{1}{|B_M|} \sum_{x \in B_M} \left(1 - \frac{\sum \text{CAM}_y(x) \odot M_{\text{dil}}(x)}{\sum \text{CAM}_y(x) + \epsilon}\right), \quad (2)$$

where B_M are the training images of the batch that have a tumor mask, M_{dil} is the mask dilated by a 31×31 kernel at 224–300 px (41×41 at 384 px), and $\lambda = 1$. The dilation keeps the peritumoral region, which Cheng et al. [2] showed to be informative. CAM_y is computed with `torch.autograd.grad(..., create_graph=True)`, so the loss is differentiable through the gradients (double back-propagation). Validation and test images are processed normally: **no mask is needed at inference**.

3.4 Tumor-region classifier

A second classifier sees a square crop centred on the tumor with side = $2 \times$ the longest side of the tumor box (at least a quarter of the image), resized to the network input. The lesion is therefore magnified about $2\text{--}5\times$. Training crops come from the ground-truth mask with random scale ($\times 0.8\text{--}1.25$) and shift ($\pm 10\%$) jitter. Validation and test crops come from the box of the *out-of-fold* Mask R-CNN of the same fold (Section 8), which never saw the validation or test patients. When nothing is detected the full image is used.

3.5 Ensemble selection and refit

Each candidate configuration is trained with three seeds (42, 1, 2). Every subset of configurations is scored by the pooled validation accuracy of the uniform average of its members’ softmax outputs. Selecting whole configurations rather than individual models is less prone to fitting validation noise: per-model greedy selection [15] reached 97.72 % validation but only 96.79 % test in our runs. The selected subset is then refit (Section 2) and averaged.

4 Explainability: Grad-CAM analysis

Table 2: Grad-CAM localisation on tumor test images (4-class runs, all folds pooled). Chance level for *energy in tumor* is 0.017. *Brain (healthy)*: CAM mass inside the head on healthy scans.

Model	Recipe	Pointing	Energy in tumor	Energy in context	CAM-IoU	Blank CAM	Brain (healthy)
AlexNet	baseline	0.160	0.029	0.069	0.052	0.000	0.735
	improved	0.119	0.026	0.058	0.047	0.000	0.495
	+ guidance	0.499	0.305	0.468	0.166	0.446	0.483
InceptionV3	baseline	0.271	0.047	0.108	0.069	0.000	0.665
	improved	0.460	0.081	0.179	0.136	0.000	0.650
	+ guidance	0.814	0.323	0.597	0.216	0.097	0.692
EfficientNet-B3	baseline	0.161	0.048	0.115	0.061	0.000	0.785
	improved	0.259	0.094	0.197	0.129	0.000	0.796
	+ guidance	0.910	0.447	0.752	0.233	0.040	0.718

Finding 1: right for the wrong reasons. Without guidance the CNNs put only 3–9 % of their Grad-CAM mass on the tumor ($1.5\text{--}5.5\times$ chance). The Grad-CAM maximum lies on the tumor in only 12–46 % of tumor images (Table 2). Visual inspection (Fig. 2a) shows correct predictions explained by the skull, the orbits, image corners or background.

Finding 2: accuracy and localisation are decoupled. The improved recipe adds about three accuracy points, but the pointing game of InceptionV3 only rises from 0.27 to 0.46 and AlexNet’s even falls.

Finding 3: a shortcut on the healthy class. On healthy scans 20–52 % of the CAM mass lies outside the head. Healthy scans come from another source and include T2 and FLAIR images, while every tumor slice is CE-T1. The near-perfect healthy-class recall (0.99–1.00) of the 4-class models therefore partly measures “does this look like a Figshare scan”. This is why the main results use the three Figshare classes.

5 Grad-CAM-guided training

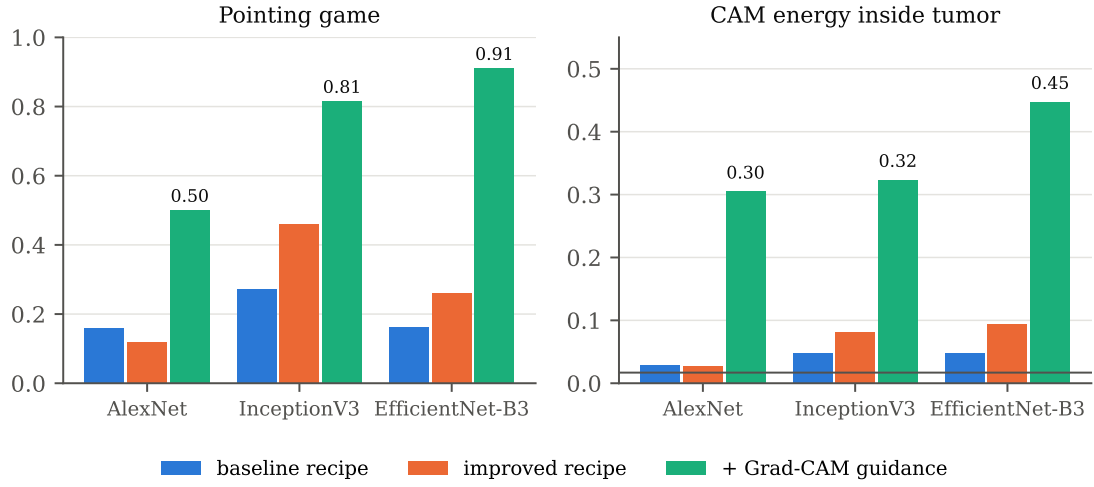
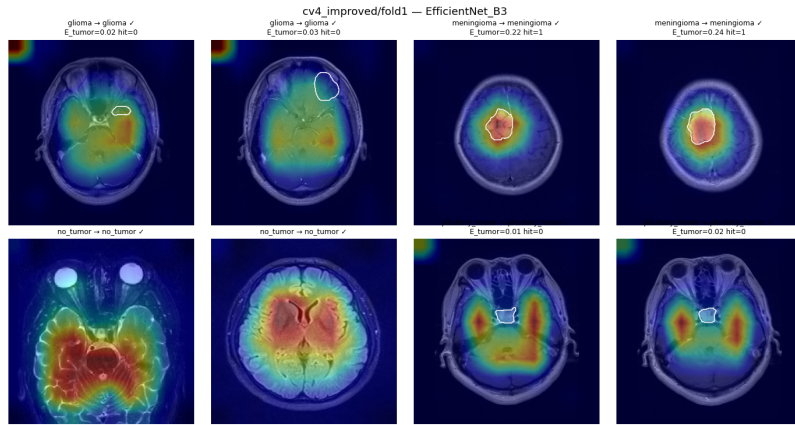
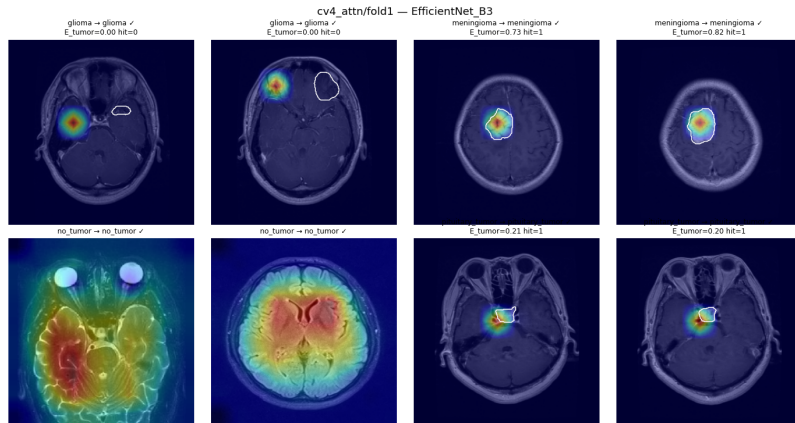


Figure 1: Pointing game and Grad-CAM energy inside the tumor (4 classes, all folds). The horizontal line in the right panel is the chance level of a uniform map (0.017).



(a) improved recipe



(b) + Grad-CAM guidance

Figure 2: EfficientNet-B3 Grad-CAM on the same held-out images (fold 1, two per class). White contour: ground-truth tumor; title: true $\rightarrow$ predicted class, energy in tumor and pointing hit.

Localisation. Guidance changes where the networks look (Table 2, Fig. 1, Fig. 2). EfficientNet-B3’s Grad-CAM maximum lands on the tumor in 91 % of tumor images (from 26 %), with 45 % of its mass on the tumor ($26\times$ chance) and 75 % within the dilated tumor region. InceptionV3 rises from 0.46 to 0.81. The final refit models keep this localisation (EfficientNet-B3: pointing 0.898, energy 0.449; InceptionV3: 0.814, 0.334).

Accuracy. On three classes, guidance improves InceptionV3 from 95.5 ± 2.2 to 96.5 ± 1.5 % and EfficientNet-B3 from 95.5 ± 1.7 to 96.7 ± 0.9 % (Fig. 3). The gain concentrates on the hardest class: meningioma recall rises from 0.895 to 0.922 (Table 5). AlexNet gains localisation (pointing $0.12 \rightarrow 0.50$) but not accuracy, and its Grad-CAM is blank on about half the images.

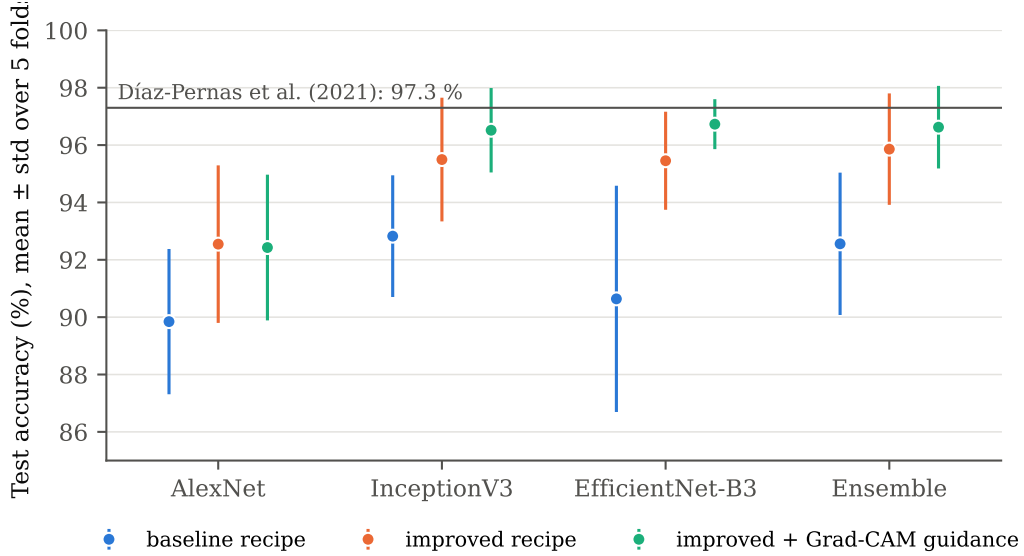


Figure 3: 3-class accuracy (3 training folds) of the baseline recipe, the improved recipe and Grad-CAM guidance; the ensemble averages AlexNet, InceptionV3 and EfficientNet-B3. The line marks Díaz-Pernas et al. [5].

Ablation: loss formulation. Our first formulation penalised $\sum \text{CAM} \odot (1 - M_{\text{dil}}) / \sum \text{CAM}$. This is identical to Eq. (2) whenever the Grad-CAM has positive mass, but it assigns zero loss to a blank Grad-CAM. AlexNet exploited this: its attention loss fell to 0.003 while all its test Grad-CAMs were blank, with unchanged accuracy (93.3 %). Writing the loss as $1 - \text{inside}$ makes a blank map the worst case. The first-generation guided InceptionV3 and EfficientNet-B3 runs (Fig. 3, Table 2) used the first form, with blank Grad-CAMs on only 4–14 % of images. All later runs (Sections 6–7) use Eq. (2).

6 Ablations

All development runs are guided, use 3 training folds and are ranked by pooled validation accuracy (Table 3).

Table 3: Development runs, 3 classes. Validation: pooled over the 5 validation folds. Test: mean $\pm$ std over the 5 test folds. Three numbers separated by slashes are seeds 42 / 1 / 2.

Configuration	Validation (%)	Test (%)
EfficientNet-B3, 300 px, no guidance (improved recipe)	–	95.45 $\pm$ 1.71
EfficientNet-B3, 300 px	97.13 / 96.93 / 96.90	96.46 / 96.35 / 96.30
EfficientNet-B3, 300 px, EMA weights (0.998)	96.34	96.19 $\pm$ 1.28
EfficientNet-B3, 448 px	–	96.46 $\pm$ 1.85
InceptionV3, 299 px	96.41 / 96.80 / 96.61	96.03 / 96.65 / 95.53
InceptionV3, 448 px	–	96.09 $\pm$ 1.02
ConvNeXt-T, 384 px, EMA	97.10 / 97.13 / 97.00	96.17 / 96.65 / 96.24
EfficientNetV2-S, 384 px, EMA	96.48 / 96.54 / 96.61	96.62 / 96.11 / 96.06
Tumor-region EfficientNet-B3, context 2 $\times$	95.20 / 95.01 / 94.35	94.42 / 93.78 / 93.94
Tumor-region EfficientNet-B3, context 3 $\times$	94.81	94.06 $\pm$ 1.93
Tumor-region ConvNeXt-T, context 2 $\times$, 384 px, EMA	94.45	94.11 $\pm$ 1.29

- **Backbone, EMA, resolution.** ConvNeXt-T and EfficientNetV2-S at 384 px, EMA weights and 448 px inputs all stay within ± 0.5 points of EfficientNet-B3 at 300 px. The bottleneck is not model capacity.
- **Seeds.** The test accuracy of one configuration varies by up to 1.1 points across seeds (InceptionV3), as large as many differences between configurations. This motivates selecting and averaging over seeds.
- **Tumor-region crops.** Alone they are 2 points weaker than full images. Small context (2 $\times$) is better than 3 $\times$, and EfficientNet-B3 is better than ConvNeXt-T. They make different errors, however (Section 7).

7 Final model

Table 4: Ensembles (uniform average, 3 seeds per configuration). The configuration subset was selected on validation (top row in bold); the refit rows are evaluated on test once.

Ensemble	Training data	Validation (%)	Test (%)
EfficientNet-B3 + InceptionV3 (6 models)	3 folds	97.26	–
+ tumor-region EfficientNet-B3 (9 models)	3 folds	97.78	97.18 $\pm$ 1.22
Refit EfficientNet-B3 (3 models)	4 folds	–	97.14 $\pm$ 0.99
Refit InceptionV3 (3 models)	4 folds	–	96.99 $\pm$ 1.17
Refit tumor-region EfficientNet-B3 (3 models)	4 folds	–	94.54 $\pm$ 0.92
Refit without tumor-region models (6 models, ablation)	4 folds	–	97.37 $\pm$ 1.05
Refit, selected ensemble (9 models)	4 folds	–	97.64 $\pm$ 1.12

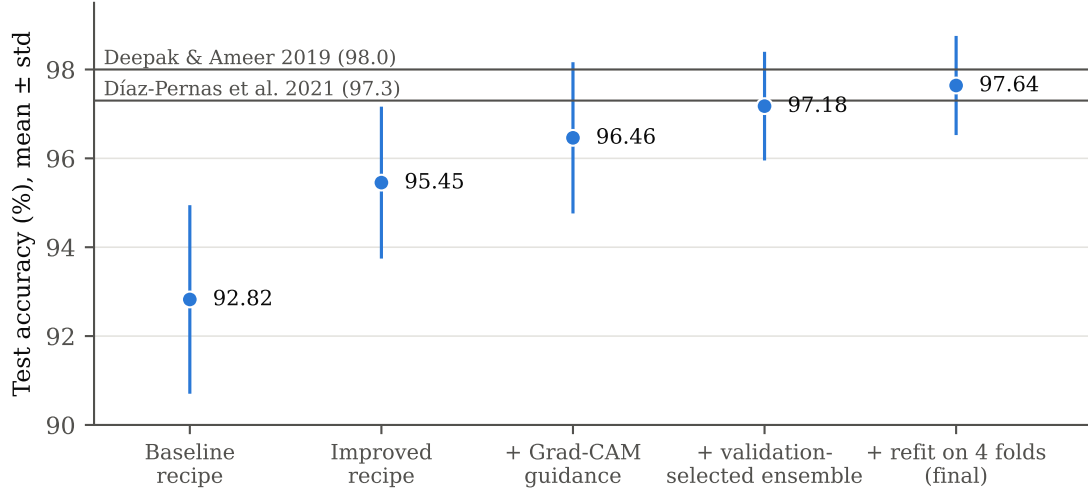


Figure 4: Test accuracy (3 classes, patient-wise 5-fold, mean $\pm$ std) along the method: baseline recipe (InceptionV3), improved recipe and Grad-CAM guidance (EfficientNet-B3), validation-selected 9-model ensemble, and its refit on four folds.

Among all $2^5 - 1$ subsets of the five configurations (EfficientNet-B3, InceptionV3, ConvNeXt-T, EfficientNetV2-S, tumor-region EfficientNet-B3), validation selects *EfficientNet-B3 + InceptionV3 + tumor-region EfficientNet-B3* (Table 4). The tumor-region models, although weaker alone, add +0.5 points on validation and +0.3 on test. Refitting on four folds adds +0.35 to +0.65 points to every configuration and +0.46 to the ensemble. The final ensemble reaches **97.64 $\pm$ 1.12 %** (pooled 97.58 %, macro-F1 97.24 %; per fold 99.26, 97.20, 97.38, 98.09, 96.27).

Table 5: Pooled per-class recall, 3 classes. Rows 1–3: EfficientNet-B3 trained on 3 folds.

Model	Glioma	Meningioma	Pituitary
baseline recipe	0.907	0.818	0.969
improved recipe	0.968	0.895	0.977
+ guidance	0.978	0.922	0.984
final ensemble	0.985	0.941	0.988
Díaz-Pernas et al. [5]	0.99	0.93	0.98

Table 6: 3-class accuracy on the Figshare dataset under patient-wise cross-validation.

Work	Method	Protocol	Accuracy
Cheng et al. 2015 [2]	tumor-region augmentation, BoW + SVM	patient-wise 5-fold	91.28 %
Anaraki et al. 2019 (as reported in [5])	CNN + genetic algorithm	5-fold	94.2 %
Díaz-Pernas et al. 2021 [5]	multiscale CNN	patient-wise 5-fold	97.3 %
Deepak & Ameer 2019 [4]	GoogLeNet features + SVM/KNN	patient-level 5-fold	98 %
This work, single EfficientNet-B3 + guidance (refit, mean of 3 seeds)	fine-tuned CNN	patient-wise 5-fold	97.01 %
This work, final ensemble	9 guided CNNs incl. tumor-region models	patient-wise 5-fold	97.64 %

The final ensemble is 0.34 points above Díaz-Pernas et al. [5], the closest published work (same dataset, protocol and classification+segmentation scope), and 0.36 points below Deepak and Ameer [4] (Table 6). Both margins are smaller than the fold-to-fold standard deviation (1.1 points), so the honest summary is *on par with the best published results*. Its meningioma recall (0.941) is the highest of the three.

8 Segmentation

Table 7: Mask R-CNN, patient-wise 5-fold CV (mean $\pm$ std). Misses count as IoU = Dice = 0; specificity is measured on the 395 healthy scans.

Threshold	Precision	Recall	Specificity	F1	Mean IoU	Mean Dice
0.4	0.967 ± 0.006	0.955 ± 0.008	0.749 ± 0.058	0.961 ± 0.002	0.682 ± 0.022	0.757 ± 0.020
0.3 (tuned on val.)	0.958 ± 0.008	0.967 ± 0.005	0.673 ± 0.075	0.963 ± 0.004	0.686 ± 0.021	0.763 ± 0.020

Mask R-CNN [11] (ResNet-50 FPN, COCO initialisation, two classes, healthy scans with empty targets) is trained per fold with SGD, cosine schedule and gradient clipping. The checkpoint is selected by validation Dice and the score threshold by validation F1 (Table 7). It detects 95–97 % of tumors with mean Dice 0.76, but a quarter to a third of healthy scans receive a false detection. Díaz-Pernas et al. [5] report Dice 0.828 with a pixel-classification CNN, so segmentation is the stage with the larger gap to the literature. Its out-of-fold boxes drive the tumor-region classifier of Section 3.4.

9 Limitations

- The margins to the published results are within one standard deviation of the fold-to-fold spread.
- Configurations and the ensemble were selected on validation folds and fixed before the refit. However, test accuracies of development candidates were computed and inspected alongside validation accuracies during the study, so the development-run test numbers are not fully blind.
- Grad-CAM at the last block is coarse (8×8 – 13×13) and reflects one layer and one explanation method.
- The 4-class setting relies on healthy scans from another source (Finding 3).
- 2D slices only; no external test set and no clinical validation.

10 Conclusion

Grad-CAM, measured against tumor masks, shows that accurate brain-tumor classifiers largely ignore the tumor. Using the masks as a training signal for the Grad-CAM fixes this (pointing game $0.26 \rightarrow 0.91$) and improves accuracy by 1.0–1.3 points with no test-time cost. Combined with tumor-region classifiers on out-of-fold Mask R-CNN crops, validation-only ensemble selection and refitting on 80 % of the data, the pipeline reaches 97.64 ± 1.12 % under the official patient-wise protocol. This exceeds the closest published result (97.3 %) and is on par with the best (98 %). Larger backbones, EMA and higher resolution did not help; better grounding, complementary views of the tumor and more training data did.

Reproducibility

All experiments are scripts in `scripts/`, run on one NVIDIA RTX PRO 5000 (48 GB) with Python 3.10, PyTorch 2.14.1, torchvision 0.29.1 and Albumentations 2.0.8. Every run stores its configuration, history and validation/test probabilities as JSON in `results/`. `scripts/select_ensemble.py`, `scripts/summarize.py` and `scripts/make_report_figures.py` regenerate every table and figure of this report from those files.

References

- [1] J. Cheng. Brain tumor dataset. figshare, dataset 1512427. <https://doi.org/10.6084/m9.figshare.1512427>
- [2] J. Cheng, W. Huang, S. Cao, R. Yang, W. Yang, Z. Yun, Z. Wang, Q. Feng. Enhanced performance of brain tumor classification via tumor region augmentation and partition. *PLoS ONE* 10(10): e0140381, 2015.
- [3] S. Bhuvaji, A. Kadam, P. Bhumkar, S. Dedge, S. Kanchan. Brain Tumor Classification (MRI). Kaggle, 2020.
- [4] S. Deepak, P. M. Ameer. Brain tumor classification using deep CNN features via transfer learning. *Computers in Biology and Medicine* 111: 103345, 2019.
- [5] F. J. Díaz-Pernas, M. Martínez-Zarzuela, M. Antón-Rodríguez, D. González-Ortega. A deep learning approach for brain tumor classification and segmentation using a multiscale convolutional neural network. *Healthcare* 9(2): 153, 2021.
- [6] A. Krizhevsky, I. Sutskever, G. E. Hinton. ImageNet classification with deep convolutional neural networks. *NeurIPS*, 2012.
- [7] C. Szegedy, V. Vanhoucke, S. Ioffe, J. Shlens, Z. Wojna. Rethinking the Inception architecture for computer vision. *CVPR*, 2016.
- [8] M. Tan, Q. V. Le. EfficientNet: Rethinking model scaling for convolutional neural networks. *ICML*, 2019.
- [9] M. Tan, Q. V. Le. EfficientNetV2: Smaller models and faster training. *ICML*, 2021.
- [10] Z. Liu, H. Mao, C.-Y. Wu, C. Feichtenhofer, T. Darrell, S. Xie. A ConvNet for the 2020s. *CVPR*, 2022.
- [11] K. He, G. Gkioxari, P. Dollár, R. Girshick. Mask R-CNN. *ICCV*, 2017.
- [12] R. R. Selvaraju, M. Cogswell, A. Das, R. Vedantam, D. Parikh, D. Batra. Grad-CAM: Visual explanations from deep networks via gradient-based localization. *ICCV*, 2017.
- [13] J. Zhang, S. A. Bargal, Z. Lin, J. Brandt, X. Shen, S. Sclaroff. Top-down neural attention by excitation backprop. *IJCV* 126: 1084–1102, 2018.
- [14] A. S. Ross, M. C. Hughes, F. Doshi-Velez. Right for the right reasons: Training differentiable models by constraining their explanations. *IJCAI*, 2017.
- [15] R. Caruana, A. Niculescu-Mizil, G. Crew, A. Ksikes. Ensemble selection from libraries of models. *ICML*, 2004.